

LK-99

LK-99 (from the **Lee-Kim 1999** research),^[2] also called **PCPOSOS**,^[3] is a gray-black, polycrystalline compound, identified as a copper-doped lead-oxyapatite. A team from Korea University led by Lee Sukbae (이석배) and Kim Ji-Hoon (김지훈) began studying this material as a potential superconductor starting in 1999.^{[4]:1} In July 2023, they published preprints claiming that it acts as a room-temperature superconductor^{[4]:8} at temperatures of up to 400 K (127 °C; 260 °F) at ambient pressure.^{[2][5][4]:1}

Many different researchers have attempted to replicate the work, and were able to reach initial results within weeks, as the process of producing the material is relatively straightforward.^[6] By mid-August 2023, the consensus^[4] was that LK-99 is not a superconductor at room temperature, and is an insulator in pure form.^{[7][8][9]}

As of 12 February 2024, no replications had gone through the peer review process of a journal, but some had been reviewed by a materials science lab. A number of replication attempts identified non-superconducting ferromagnetic and diamagnetic causes for observations that suggested superconductivity. A prominent cause was a copper sulfide impurity^[10] occurring during the proposed synthesis, which can produce resistance drops, lambda transition in heat capacity, and magnetic response in small samples.^{[11][12][10][13][14][15][16]}

After the initial preprints were published, Lee claimed they were incomplete,^[17] and coauthor Kim Hyun-Tak (김현탁) said one of the papers contained flaws.^[18]

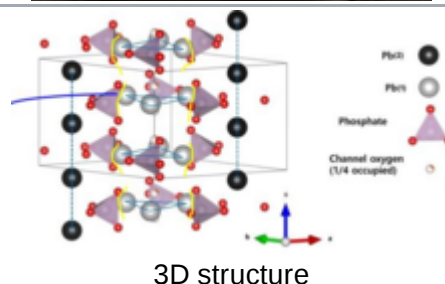
Chemical properties and structure

The chemical composition of LK-99 is approximately $\text{Pb}_9\text{Cu}(\text{PO}_4)_6\text{O}$, in which—compared to pure lead-apatite ($\text{Pb}_{10}(\text{PO}_4)_6\text{O}$)^{[19]:5}—approximately one quarter of Pb(II) ions in position 2 of the apatite structure are replaced by Cu(II) ions.^{[4]:9}

The structure is similar to that of apatite, space group $P6_3/m$ (No. 176).

Synthesis

LK-99



Identifiers

InChI

InChI=1S/Cu.6H3O4P.O.9Pb/c;6*1-5(2,3)4;;;;;;;;;/h;6*(H3,1,2,3,4);;;;;;;;;/q+2;;;;;;;;;-2;9*+2/p-18
Key: KZSIWLDFTIMUEG-UHFFFA
OYSA-A

SMILES

[Pb+2].[Pb+2].[Pb+2].[Pb+2].[Pb+2].
[Pb+2].[Pb+2].[Pb+2].[Pb+2].[Cu
+2].O=P([O-])([O-])[O-].O=P([O
-])([O-])[O-].O=P([O-])([O-])[O-].
O=P([O-])([O-])[O-].O=P([O-])([O
-])[O-].O=P([O-])([O-])[O-].[O-2]

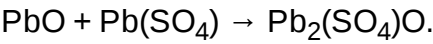
Properties

<u>Chemical formula</u>	CuO ₂₅ P ₆ Pb ₉
<u>Molar mass</u>	2 514.2 g·mol ⁻¹
<u>Appearance</u>	Purple crystal when pure ^[1]
<u>Density</u>	≈6.699 g/cm ³

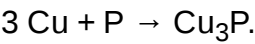
Structure

<u>Crystal structure</u>	hexagonal
<u>Space group</u>	$P6_3/m$, No. 176

Lee *et al.* provide a method for chemical synthesis of LK-99^{[19]:2} in three steps. First they produce lanarkite from a 1:1 molar mixing of lead(II) oxide (PbO) and lead(II) sulfate (Pb(SO₄)) powders, and heating at 725 °C (1,000 K; 1,340 °F) for 24 hours:



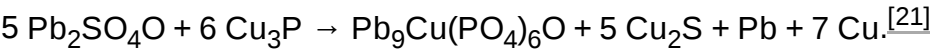
Then, copper(I) phosphide (Cu₃P) is produced by mixing copper (Cu) and phosphorus (P) powders in a 3:1 molar ratio in a sealed tube under a vacuum and heated to 550 °C (820 K; 1,000 °F) for 48 hours:^{[19]:3}



Then, lanarkite and copper phosphide crystals are ground into a powder, placed in a sealed tube under a vacuum, and heated to 925 °C (1,200 K; 1,700 °F) for between 5–20 hours:^{[19]:3}



There were a number of problems with the above synthesis from the initial paper. The reaction is not balanced, and others reported the presence of copper(I) sulfide (Cu₂S) as well.^{[20][12]} For ***x* = 1** a balanced reaction might be:



Many syntheses produced fragmentary results in different phases, where some of the resulting fragments were responsive to magnetic fields, other fragments were not.^[22] The first synthesis to produce pure crystals found them to be diamagnetic insulators.^[23]

Physical properties

Some small LK-99 samples were reported to show strong diamagnetic properties, including a response confusingly^[24] referred to as "partial levitation" over a magnet.^[19] This was misinterpreted by some as a sign of superconductivity, although it is a sign of regular diamagnetism or ferromagnetism.

While initial preprints claimed the material was a room-temperature superconductor,^{[19]:1} they did not report observing any definitive features of superconductivity, such as zero resistance, the Meissner effect, flux pinning, AC magnetic susceptibility, the Josephson effect, a temperature-dependent critical field and current, or a sudden jump in specific heat around the critical temperature.^[25]

As it is common for a new material to spuriously seem like a potential candidate for high-temperature superconductivity,^[14] thorough experimental reports normally demonstrate a number of these expected properties. As of 15 October 2023, not one of these properties had been observed by the original experiment or any replications.^[26]

Proposed mechanism for superconductivity

<u>Lattice constant</u>	<i>a</i> = 9.843 Å, <i>c</i> = 7.428 Å
<u>Lattice volume (V)</u>	623.2 Å ³
<u>Formula units (Z)</u>	1
Related compounds	
<u>Related compounds</u>	<u>Oxyppyromorphite</u> (lead apatite)
Except where otherwise noted, data are given for materials in their <u>standard state</u> (at 25 °C [77 °F], 100 kPa).	
<u>Infobox references</u>	

Partial replacement of Pb^{2+} ions with smaller Cu^{2+} ions is said to cause a 0.48% reduction in volume, creating internal stress in the material,^{[4]:8} causing a heterojunction quantum well between the $\text{Pb}(1)$ and oxygen within the phosphate ($[\text{PO}_4]^{3-}$). This quantum well was proposed to be superconducting^{[4]:10}, based on a 2021 paper^[27] by Kim Hyun-Tak describing a novel and complicated theory combining ideas from a classical theory of metal-insulator transitions,^[28] the standard Bardeen–Cooper–Schrieffer theory, and the theory of hole superconductivity^[29] by J.E.Hirsch.

Response

On 31 July 2023, Sinéad Griffin of Lawrence Berkeley National Laboratory analyzed LK-99 with density functional theory (DFT), showing that its structure would have correlated isolated flat bands, and suggesting this might contribute to superconductivity.^[30] However, while other researchers agreed with the DFT analysis, a number suggested that this was not compatible with superconductivity, and that a structure different from what was described in Lee, *et al.* would be necessary.^[31]

Analyses by industrial and experimental physicists noted experimental and theoretical shortcomings of the published works.^[32] Shortcomings included the lack of phase diagrams^[29] spanning temperature, stoichiometry,^[33] and stress; the lack of pathways for the very high T_c of LK-99 compared to prior heavy fermion superconductors; the absence of flux pinning in any observations; the possibility of stochastic conductive artifacts^[34] in conductivity measurements; the high resistance and low current capacity of the alleged superconducting state; and the lack of direct transmission electron microscopy (TEM) of the materials.

Compound name

The name LK-99 comes from the initials of discoverers Lee and Kim, and the year of discovery (1999).^[2] The pair had worked with Tong-Seek Chair (최동식) at Korea University in the 1990s.^[35]

In 2008, they founded the Quantum Energy Research Centre (퀀텀 에너지연구소; also known as Q-Centre) with other researchers from Korea University.^[17] Lee would later become CEO of Q-Centre, and Kim would become director of research and development.

Publication history

Lee has stated that in 2020, an initial paper was submitted to Nature, but was rejected.^[35] Similarly presented research on room-temperature superconductors (but a completely different chemical system) by Ranga P. Dias had been published in Nature earlier that year, and received with skepticism—Dias's paper would subsequently be retracted in 2022 after its data was questioned as having been falsified.^[36]

In 2020, Lee and Kim Ji-Hoon filed a patent application.^[37] A second patent application (additionally listing Young-Wan Kwon), was filed in 2021, which was published on 3 March 2023.^[38] A World Intellectual Property Organization (WIPO) patent was also published on 2 March 2023.^[39] On 4 April 2023, a Korean trademark application for "LK-99" was filed by the Q-Centre.^[40]

Scholarly articles and preprints

A series of academic publications summarizing initial findings came out in 2023, with a total of seven authors across four publications.

On 31 March 2023, a Korean-language paper, "Consideration for the development of room-temperature ambient-pressure superconductor (LK-99)", was submitted to the *Korean Journal of Crystal Growth and Crystal Technology*.^[5] It was accepted on 18 April, but was not widely read until three months later.

On 22 July 2023, two preprints appeared on arXiv. The first was submitted by Young-Wan Kwon, and listed Kwon, former Q-Centre CTO, as third author.^[4] The second preprint was submitted only 2 hours later by Kim Hyun-Tak, former principal researcher at the Electronics & Telecommunications Research Institute and professor at the College of William & Mary, listing himself as third author, as well as three new authors.^{[19][41]}

On 23 July, the findings were also submitted by Lee to *APL Materials* for peer review.^{[35][17]} On 3 August 2023, a newly-formed Korean LK-99 Verification Committee requested a high-quality sample from the original research team. The team responded that they would only provide the sample once the review process of their APL paper was completed, expected to take several weeks or months.^[42]

On 31 July 2023, a group led by Kapil Kumar published a preprint on arXiv documenting their replication attempts, which confirmed the structure using X-ray crystallography (XRD) but failed to find strong diamagnetism.^[20]

On 16 August 2023, *Nature* published an article declaring that LK-99 had been demonstrated to not be a superconductor, but rather an insulator. It cited statements by an condensed matter experimentalist at the University of California, Davis, and several studies previewed in August 2023.^[1]

Other discussion by authors

On 26 July 2023, Kim Hyun-Tak stated in an interview with the *New Scientist* that the first paper submitted by Kwon contained "many defects" and was submitted without his permission.^{[33][41]}

On 28 July 2023, Kwon presented the findings at a symposium held at Korea University.^{[43][44][45]} That same day, Yonhap News Agency published an article quoting an official from Korea University as saying that Kwon was no longer in contact with the university.^[17] The article also quoted Lee saying that Kwon had left the Q-Centre Research Institute four months previously.^[17]

On the same day, Kim Hyun-Tak provided *The New York Times* with a new video presumably showing a sample displaying strong signs of diamagnetism.^[2] The video (https://vp.nyt.com/video/2023/08/03/110377_1_03superconductor-vid_wg_1080p.mp4) appears to show a sample different to the one in the original preprint. On 4 August 2023, he informed SBS News that high-quality LK-99 samples may exhibit diamagnetism over 5,000 times greater than graphite, which he claimed would be inexplicable unless the substance is a superconductor.^[46]

Response

Materials scientists and superconductor researchers responded with skepticism.^{[18][47]} The highest-temperature superconductors known at the time of publication had a critical temperature of 250 K (−23 °C; −10 °F) at pressures of over 170 gigapascals (1,680,000 atm; 24,700,000 psi). The highest-temperature superconductors at atmospheric pressure (1 atm) had a critical temperature of at most 150 K (−123 °C; −190 °F).

On 2 August 2023, the *The Korean Society of Superconductivity and Cryogenics* established a verification committee as a response to the controversy and unverified claims of LK-99, in order to arrive at conclusions over these claims. The verification committee is headed by Kim Chang-Young of Seoul National University and consists of members of the university, Sungkyunkwan University and Pohang University of Science and Technology. Upon formation, the verification committee did not agree that the two 22 July arXiv papers by Lee *et al.* or the publicly available videos at the time supported the claim of LK-99 being a superconductor.^{[41][48]}

As of 15 August 2023, the measured properties do not prove that LK-99 is a superconductor. The published material does not explain how the LK-99's magnetisation can change, demonstrate its specific heat capacity, or demonstrate it crossing its transition temperature.^[18] A more likely explanation for LK-99's magnetic response is a mix of ferromagnetism and non-superconductive diamagnetism.^{[41][16][49]} A number of studies found that copper(I) sulfide contamination common to the synthesis process could closely replicate the observations that inspired the initial preprints.^{[10][11]}

Public response

The claims in the 22 July papers by Lee *et al.* went viral on social media platforms the following week.^{[6][50]} The viral nature of the claim resulted in posts from users using pseudonyms from Russia and China claiming to have replicated LK-99 on both Twitter and Zhihu.^[51] Other viral videos described themselves as having replicated samples of LK-99 "partially levitating", most of which were found to be fake.^[47]

Scientists interviewed by the press remained skeptical,^{[52][53]} because of the quality of both the original preprints, the lack of purity in the sample they reported, and the legitimacy of the claim after the failure of previous claims of room temperature superconductivity did not show legitimacy (such as the Ranga Dias affair).^[41] The *Korean Society of Superconductivity and Cryogenics* expressed concern on the social and economic impacts of the preliminary and unverified LK-99 research.^[54]

A video from Huazhong University of Science and Technology uploaded on 1 August 2023 by a postdoctoral researcher on the team of Chang Haixin,^[41] apparently showed a micrometre-sized sample of LK-99 partially levitating. This went viral on Chinese social media, becoming the most viewed video on Bilibili by the next day,^{[55][41]} and a prediction market briefly put the chance of successful replication at 60%.^[56] A researcher from the Chinese Academy of Sciences refused to comment on the video for the press, dismissing the claim as "ridiculous".^[55]

In early August, people began to create memes about "floating rocks",^[57] and there was a brief surge in Korean and Chinese technology stocks,^{[58][59][60]} despite warnings from the Korean stock exchange against speculative bets in light of the excitement around LK-99,^[54] which eventually fell on August 8.^[61] Following the publication of the *Nature* article on August 16 that proclaimed LK-99 is not a superconductor,^[1] South Korean superconductor stocks fell further, as the interest about LK-99 from investors in previous weeks disappeared.^[62]

Replication attempts

After the July 2023 publication's release, independent groups reported that they had begun attempting to reproduce the synthesis, with initial results expected within weeks.^[6]

As of 15 August 2023, no replication attempts had yet been peer-reviewed by a journal. Of the non-peer-reviewed attempts, over 15 notable labs have published results that failed to observe any superconductivity, and a few have observed magnetic response in small fragments that could be explained by normal diamagnetism or ferromagnetism. Some demonstrated and replicated alternate causes of the observations in the original papers: Copper-deficient copper (I) sulfide^[10] has a known phase transition at 377 K (104 °C; 219 °F) from a low-temperature phase to a high-temperature superionic phase, with a sharp rise in resistivity^{[11][10]} and a λ -like-feature in the heat capacity.^[10] Furthermore, Cu₂S is diamagnetic.

Only one attempt observed any sign of superconductivity: Southeast University claimed to measure very low resistance in a flake of LK-99, in one of four synthesis attempts, below a temperature of 110 K (−163 °C; −262 °F).^{[2][63]} Doubts were expressed by experts in the field, as they saw no dropoff to zero resistance, and used crude instruments that could not measure resistance below 10 $\mu\Omega$ (too high to distinguish superconductivity from less exotic low-temperature conductivity), and had large measurement artifacts.^{[47][64]}

Some replication efforts gained global visibility, with the aid of online replication trackers that catalogued new announcements and status updates.^{[51][26]}

Experimental studies

Selected experimental studies.

Results Key:  Success  Partial success  Partial failure  Failure

Group	Country/region	Status	Results	Publication notes
<u>Max Planck (Solid State)</u>	 <u>Germany</u>	Preliminary	Produced pure LK-99 samples with floating zone technique. Purple crystals with high resistance, no magnetic response.	arXiv: ^[65] Press: ^{[1][66]}
<u>Huazhong Tech</u>	 <u>China</u>	Preliminary	Measured diamagnetism of micron-sized flakes. Non-zero resistance, purity of sample was important.	arXiv: ^{[32][15]} bilibili video: ^{[34][67]} Press: ^{[26][41][55][51]}
<u>Beihang University</u>		Preliminary	No diamagnetism observed. High resistivity not consistent with superconductivity.	arXiv: ^[68] Press: ^{[57][69]}
<u>Southeast University</u>		Preliminary	Structure confirmed by XRD. Resistance of one mm-sized sample dropped from 0.1 Ω at room temperature to noise level ($10^{-5} \Omega$) at 110 K and below. No observed Meissner effect.	arXiv: ^{[70][71]} Video: ^[63] Critics: ^{[47][64]} Press: ^[2]
<u>Peking University</u>		Preliminary	No Meissner effect nor zero resistivity observed.	arXiv: ^[72] Press: ^[1]
<u>Chinese Academy of Sciences (Condensed Matter)</u>		Preliminary	No superconductivity observed. Proposed that resistivity drop and strong diamagnetism could be due to a phase change of <u>Cu₂S</u> impurities.	arXiv: ^[11] Press: ^{[1][66]}
<u>Central South University, South China Tech, and UESTC</u>	 <u>Spain</u> ,  <u>USA</u> ,  <u>Germany</u>	Preliminary	Low-field microwave absorption below 250 K resembles superconductivity, but is destroyed by rotation in an external field. Theoretical models suggest the external field excites a fragile superconducting state to a vortex glass, followed by a ~2-day-long relaxation to the ground state.	arXiv: ^[73]
<u>DIPC, Princeton, Max Planck (Chemical Physics)</u>		Preliminary	Synthesized LK-99 found to be a multiphase material. Performed single-crystal analysis with XRD. Tested four different Cu dopings, some found to be magnetic but none was superconducting.	arXiv: ^[74] Press: ^{[1][66]}
<u>University of Manchester</u>	 <u>United Kingdom</u>	Preliminary	Synthesized and characterized samples of LK-99, no superconductivity.	arXiv: ^[75]
<u>CSIR-NPLI</u>	 <u>India</u>	Preliminary	Initial attempt: Structure confirmed by XRD, no diamagnetism or superconductivity.	Video: ^[76] arXiv: ^{[20][77][78]} Press: ^{[41][57][51][69]}

			Second attempt: strong diamagnetism in a fragment.	
<u>Varda Space & USC</u>	 <u>United States</u>	Preliminary	Only a few LK-99 fragments responded to magnetic field. Analysis showed impurities of Iron and Cu ₂ S, which could explain magnetic response rather than superconductivity.	Videos: [79] [80] [81] [13] Press: [66] [41] [51] [50]
<u>UC–Boulder</u>		Unpublished	Samples have failed tests for superconductivity.	Press: [7]
<u>Argonne</u>		Unknown	Not reported	Press: [41] [82] [51]
<u>Korea University, Sungkyunkwan University, Seoul National University</u>	 <u>South Korea</u>	Unknown	Not reported	Press: [41] [48]
<u>Chinese Academy of Sciences (Process Engineering), South China Tech, Beijing 2060, Huazhong Tech, Fuzhou University, Tokai University, and USTB</u>	 <u>Mainland China</u> ,  <u>Japan</u>	Preliminary	Modified LK-99 exhibited diamagnetic direct current magnetization occurred under a 25 Oe magnetic field, but significant bifurcation between zero field cooling (ZFC) and field cooling (FC) measurements, and paramagnetism at a 200 Oe magnetic field. A glassy memory effect was discovered while cooling. Typical hysteresis loops of superconductors were detected below 250 K, and there was asymmetry between forward and reverse magnetic field scans. Possible Meissner effect at room temperature.	Press: [83] [84] arXiv: [85] Q&As: [86] [87] [88]
<u>Chinese Academy of Sciences (Process Engineering), Huazhong University of Science and Technology, University of Science and Technology Beijing, South China University of Technology, Fuzhou University, Tokai University and University of Science and Technology of China</u>		Preliminary	1. Proposed a new Lk99 structure theory 2. The resistance of Lk99 material was measured, which is roughly equivalent to copper. 3. Observed <u>strange metal phenomena</u>	arXiv: <u>Observation of diamagnetic strange-metal phase in sulfur-copper codoped lead apatite</u> (https://arxiv.org/html/2403.11126v1)

Theoretical studies

In the initial papers, the theoretical explanations for potential mechanisms of superconductivity in LK-99 were incomplete. Later analyses by other labs added simulations and theoretical evaluations of the material's electronic properties from first principles.

Selected theoretical studies:

Group	Country	Result	Publication notes
Chinese Academy of Sciences (SYNL)	 China	First-principles study of the electronic structure of LK-99 and other variants. Expresses no opinion on room-temp superconductivity.	arXiv: Junwen Lai, <i>et al.</i> ^[89] Media mentions: ^[90]
Lawrence Berkeley National Laboratory	 United States	Density functional theory analysis on a simplified 3D structure explored possible electronic structure that could favor superconductivity, suggests slightly decreased lattice constant. Similar work published the next day by Si & Held ^[31] and Kurlito, <i>et al.</i> ^[91]	arXiv: Sinéad Griffin ^{[30][b 1]} Analysis: ^{[92][93]} Media mentions: ^{[56][57]}
Universidad de Chile	 Chile	DFT analysis, finding large electron-phonon coupling in the flat bands.	arXiv: J. Cabezas-Escares, <i>et al.</i> ^[94]
CIEMAT	 Spain ,  Armenia	Concludes the original synthesis for LK-99 likely produces a heterogenous material, making it hard for others to reproduce the same results	arXiv: P. Abramian, <i>et al.</i> ^[22]
Northwest University (China) and TU Wien	 China ,  Austria	Concludes $\text{Pb}_9\text{Cu}(\text{PO}_4)_6\text{O}$, without further doping, is an insulator. Analyzes possible effects of doping.	arXiv: Liang Si & Karsten Held ^{[31][b 1]}
Indiana University Bloomington	 United States	Concludes material is a transparent insulator, possibly with active Cu color centers at low temperature. Does not find signatures of type I or II superconductivity. Solves previous issues related to overestimation of lattice constant contraction, doping site energetics. Does not find flat bands at Fermi level, concluding they are related to an unfavored high-symmetry structure.	arxiv: A.B. Georgescu (https://georgescu.lab.indiana.edu/) ^[95] Analysis and discussions: ^{[96][97]}

1. The first three **density functional theory** analyses were published within 24 hours of one another, and have largely overlapping analysis.

See also

- [Bismuth strontium calcium copper oxide](#): Superconductivity at $T_c \approx 33 \text{ K}$ (-240.2 °C) to 104 K (-169 °C)
- [Carbonaceous sulfur hydride](#): Purported superconductivity at $T_c \approx 288 \text{ K}$ (15 °C) at 267 GPa
- [Lanthanum decahydride](#): Superconductivity at $T_c = 250 \text{ K}$ (-23 °C) at 150 GPa
- [Unconventional superconductor](#)

- Salvatore Pais – Inventor with a patent referenced by a patent related to LK-99

References

1. Garisto, Dan (16 August 2023). "LK-99 isn't a superconductor — how science sleuths solved the mystery: Replications pieced together the puzzle of why the material displayed superconducting-like behaviours" (<https://www.nature.com/articles/d41586-023-02585-7>). *Nature*. **620** (7975): 705–706. doi:10.1038/d41586-023-02585-7 (<https://doi.org/10.1038%2Fd41586-023-02585-7>). PMID 37587284 (<https://pubmed.ncbi.nlm.nih.gov/37587284>). S2CID 260955242 (<https://api.semanticscholar.org/CorpusID:260955242>). Archived (<https://web.archive.org/web/20230817012222/https://www.nature.com/articles/d41586-023-02585-7>) from the original on 17 August 2023. Retrieved 17 August 2023.
2. Chang, Kenneth (3 August 2023). "LK-99 Is the Superconductor of the Summer" (<https://www.nytimes.com/2023/08/03/science/lk-99-superconductor-ambient.html>). *The New York Times*. Archived (<https://web.archive.org/web/20230803183814/https://www.nytimes.com/2023/08/03/science/lk-99-superconductor-ambient.html>) from the original on 3 August 2023. Retrieved 3 August 2023.
3. Bulletin of the American Physical Society (<https://meetings.aps.org/Meeting/MAR24/Session/A16.2>), March 2024 meeting notice
4. Lee, Sukbae; Kim, Ji-Hoon; Kwon, Young-Wan (22 July 2023). "The First Room-Temperature Ambient-Pressure Superconductor". arXiv:2307.12008 (<https://arxiv.org/abs/2307.12008>) [[cond-mat.supr-con](https://arxiv.org/archive/cond-mat/supr-con) (<https://arxiv.org/archive/cond-mat/supr-con>)].
5. Lee, Sukbae; Kim, Ji-Hoon; Im, Sungyeon; An, Soomin; Kwon, Young-Wan; Auh, Keun Ho (31 March 2023). "Consideration for the development of room-temperature ambient-pressure superconductor (LK-99)" (<http://journal.kci.go.kr/jkcgct/archive/articleView?artId=ART002955269>). *Korean Crystal Growth and Crystal Technology*. **33** (2). Korea Association Of Crystal Growth: 61–70. doi:10.6111/JKCGCT.2023.33.2.061 (<https://doi.org/10.6111%2FJKCGCT.2023.33.2.061>). Archived (<https://web.archive.org/web/20230725211223/http://journal.kci.go.kr/jkcgct/archive/articleView?artId=ART002955269>) from the original on 25 July 2023. Retrieved 25 July 2023.
6. Garisto, Dan (27 July 2023). "Viral New Superconductivity Claims Leave Many Scientists Skeptical" (<https://www.scientificamerican.com/article/viral-new-superconductivity-claims-leave-many-scientists-skeptical/>). Materials science. *Scientific American*. Archived (<https://web.archive.org/web/20230727210945/https://www.scientificamerican.com/article/viral-new-superconductivity-claims-leave-many-scientists-skeptical/>) from the original on 27 July 2023. Retrieved 28 July 2023.
7. Johnson, Carolyn Y. (9 August 2023). "A superconductor claim blew up online. Science has punctured it" (<https://www.washingtonpost.com/science/2023/08/09/room-temperature-superconductor-lk99-evidence/>). *The Washington Post*. Archived (<https://web.archive.org/web/20230809215311/https://www.washingtonpost.com/science/2023/08/09/room-temperature-superconductor-lk99-evidence/>) from the original on 9 August 2023. Retrieved 9 August 2023.
8. Robinson, Dan. "LK-99 slammed as 'not a superconductor at all'" (https://www.theregister.com/2023/08/09/lk99_more_physicists_fail_reproduce/). *www.theregister.com*. Archived (https://web.archive.org/web/20230810063704/https://www.theregister.com/2023/08/09/lk99_more_physicists_fail_reproduce/) from the original on 10 August 2023. Retrieved 10 August 2023.

9. Padavic-Callaghan, Karmela. "LK-99: Mounting evidence suggests material is not a superconductor" (<https://www.newscientist.com/article/2386866-lk-99-mounting-evidence-suggests-material-is-not-a-superconductor/>). *New Scientist*. Archived (<https://web.archive.org/web/20230809230754/https://www.newscientist.com/article/2386866-lk-99-mounting-evidence-suggests-material-is-not-a-superconductor/>) from the original on 9 August 2023. Retrieved 10 August 2023.
10. Jain, Prashant K. (2023). "Phase transition of copper (I) sulfide and its implication for purported superconductivity of LK-99". arXiv:2308.05222v1 (<https://arxiv.org/abs/2308.05222v1>) [cond-mat.supr-con (<https://arxiv.org/archive/cond-mat.supr-con>)].
11. Shilin Zhu; Wei Wu; Zheng Li; Jianlin Luo (8 August 2023). "First-order transition in LK-99 containing Cu₂S". *Matter*. **6** (12): 4401–4407. arXiv:2308.04353 (<https://arxiv.org/abs/2308.04353>). doi:10.1016/j.matt.2023.11.001 (<https://doi.org/10.1016%2Fj.matt.2023.11.001>).
12. Guo, Kaizhen; Li, Yuan; Jia, Shuang (6 August 2023). "Ferromagnetic half levitation of LK-99-like synthetic samples". *Science China Physics, Mechanics & Astronomy*. **66** (10). arXiv:2308.03110 (<https://arxiv.org/abs/2308.03110>). Bibcode:2023SCPMA..6607411G (<https://ui.adsabs.harvard.edu/abs/2023SCPMA..6607411G>). doi:10.1007/s11433-023-2201-9 (<https://doi.org/10.1007%2Fs11433-023-2201-9>). S2CID 260680385 (<https://api.semanticscholar.org/CorpusID:260680385>).
13. @andrewmccalip (10 August 2023). "Meissner Effect or Bust: Day 12" (<https://x.com/andrewmccalip/status/1689476909208600576>) (Tweet) – via Twitter.
14. Fuhrer, Michael S. [@MichaelSFuhrer] (2 August 2023). "You'd think superconductivity would be easy to detect; it comes with zero electrical resistance, so if you measure resistance, and it's zero, you're done. Unfortunately there are many ways to get fooled" (<https://x.com/MichaelSFuhrer/status/1686644235414720512>) (Tweet). Retrieved 2 August 2023 – via Twitter.
15. Hao Wu; Li Yang; Jie Yu; Gaojie Zhang; Bichen Xiao; Haixin Chang (9 August 2023). "Observation of abnormal resistance-temperature behavior along with diamagnetic transition in Pb_{10-x}Cu_x(PO₄)₆O-based composite". arXiv:2308.05001 (<https://arxiv.org/abs/2308.05001>) [cond-mat.supr-con (<https://arxiv.org/archive/cond-mat.supr-con>)].
16. Orf, Darren (9 August 2023). "Well, Seems Like LK-99 Isn't a Room Temperature Superconductor After All" (<https://www.popularmechanics.com/science/green-tech/a44764458/lk-99-not-a-superconductor/>). *Popular Mechanics*. Archived (<https://web.archive.org/web/20230810010424/https://www.popularmechanics.com/science/green-tech/a44764458/lk-99-not-a-superconductor/>) from the original on 10 August 2023. Retrieved 10 August 2023.
17. 조승한 (28 July 2023). 강의영 (ed.). '상온 초전도체 구현' 한국 연구에 국내외 논란..."검증 거쳐야" (<https://www.yna.co.kr/view/AKR20230728146700017>) [Controversy both domestic and abroad regarding Korean development of room temperature superconductor ... "It has to be verified"] (in Korean). Yonhap News Agency. Archived (<https://web.archive.org/web/20230728124520/https://www.yna.co.kr/view/AKR20230728146700017>) from the original on 28 July 2023. Retrieved 28 July 2023. ... 논문이 아니며 공개도 의도한 바가 아니라고 선을 그었다. ... 이 대표는 이날 연합뉴스와 통화에서 "다른 저자들의 허락 없이 권 연구교수가 임의로 아카이브에 게재한 것"이라며 "아카이브에 내려달라는 요청을 해둔 상황"이라고 주장했다. ... 이 대표는 권 연구교수가 쿼텀에너지연구소 최고기술책임자(CTO)로 있었지만 4개월 전 이사직을 내려놓고 현재는 회사와 관련이 없다고도 밝혔다. ... 고려대 관계자에 따르면 권 연구교수는 현재 학교와도 연락이 닿지 않는 상황으로 알려졌다.

18. Padavic-Callaghan, Karmela (26 July 2023). "Room-temperature superconductor 'breakthrough' met with scepticism" (<https://www.newscientist.com/article/2384782-room-temperature-superconductor-breakthrough-met-with-scepticism/>). *New Scientist*. Archived (<https://web.archive.org/web/20230726181825/https://www.newscientist.com/article/2384782-room-temperature-superconductor-breakthrough-met-with-scepticism/>) from the original on 26 July 2023. Retrieved 26 July 2023. "Speaking to *New Scientist*, Hyun-Tak Kim at the College of William & Mary in Virginia says he will support anyone trying to replicate his team's work. ... [HT] Kim has only co-authored one of the arXiv papers, while the other is authored by his colleagues at the Quantum Energy Research Centre in South Korea, ... Both papers present similar measurements, however [HT] Kim says that the second [3-author] paper contains "many defects" and was uploaded to arXiv without his permission. ... Once the findings are published in a peer-reviewed journal, ... [HT] Kim says ... he will support anyone who wants to create and test LK-99"
19. Lee, Sukbae; Kim, Ji-Hoon; Kim, Hyun-Tak; Im, Sungyeon; An, SooMin; Auh, Keun Ho (22 July 2023). "Superconductor $\text{Pb}_{10-x}\text{Cu}_x(\text{PO}_4)_6\text{O}$ showing levitation at room temperature and atmospheric pressure and mechanism". arXiv:2307.12037 (<https://arxiv.org/abs/2307.12037>) [cond-mat.supr-con (<https://arxiv.org/archive/cond-mat.supr-con>)].
20. Kumar, Kapil; Karn, N.K.; Awana, V.P.S. (31 July 2023). "Synthesis of possible room temperature superconductor LK-99: $\text{Pb}_9\text{Cu}(\text{PO}_4)_6\text{O}$ ". *Superconductor Science and Technology*. **36** (10): 10LT02. arXiv:2307.16402 (<https://arxiv.org/abs/2307.16402>). Bibcode:2023SuScT...36jLT02K (<https://ui.adsabs.harvard.edu/abs/2023SuScT...36jLT02K>). doi:10.1088/1361-6668/acf002 (<https://doi.org/10.1088/1361-6668/acf002>). S2CID 260333984 (<https://api.semanticscholar.org/CorpusID:260333984>).
21. Kumar, Kapil; Karn, N. K.; Kumar, Yogesh; Awana, V. P. S. (7 August 2023). "Absence of superconductivity in LK-99 at ambient conditions". arXiv:2308.03544 (<https://arxiv.org/abs/2308.03544>) [cond-mat.supr-con (<https://arxiv.org/archive/cond-mat.supr-con>)].
22. Abramian, P.; Kuzanyan, A.; Nikoghosyan, V.; Teknowijoyo, S.; Gulian, A. (2023). "Some remarks on possible superconductivity of composition $\text{Pb}_9\text{CuP}_6\text{O}_{25}$ ". arXiv:2308.01723 (<https://arxiv.org/abs/2308.01723>) [cond-mat.supr-con (<https://arxiv.org/archive/cond-mat.supr-con>)].
23. Puphal, P.; Akbar, M. Y. P.; Hepting, M.; Goering, E.; Isobe, M.; Nugroho, A. A.; Keimer, B. (11 August 2023). "Single crystal synthesis, structure, and magnetism of $\text{Pb}_{10-x}\text{Cu}_x(\text{PO}_4)_6\text{O}$ ". *APL Materials*. **11** (10). arXiv:2308.06256 (<https://arxiv.org/abs/2308.06256>). Bibcode:2023APLM...11j1128P (<https://ui.adsabs.harvard.edu/abs/2023APLM...11j1128P>). doi:10.1063/5.0172755 (<https://doi.org/10.1063/5.0172755>). S2CID 260866146 (<https://api.semanticscholar.org/CorpusID:260866146>).
24. Chen, Yan-Cong (2023). "Magical or magnetic? Less commonly taught facts about real-world permanent magnets and their diverse interactions with objects". arXiv:2308.11542 (<https://arxiv.org/abs/2308.11542>) [physics.pop-ph (<https://arxiv.org/archive/physics.pop-ph>)].
25. Fuhrer, Michael S. [@MichaelSFuhrer] (2 August 2023). "So generally you'll see multiple pieces of evidence for superconductivity in a new report: Meissner effect, AC susceptibility, temperature-dependent critical field and critical current, single-particle tunnelling gap, jump in specific heat at T_c , Josephson tunnelling... etc" (<https://x.com/MichaelSFuhrer/status/1686644237742526464>) (Tweet). Retrieved 2 August 2023 – via Twitter.
26. Lowe, Derek (1 August 2023). "A Room-Temperature Superconductor? New Developments" (<https://www.science.org/content/blog-post/room-temperature-superconductor-new-developments>). *Chemical News*. *In the pipeline* (blog). American Association for the Advancement of Science. Archived (<https://web.archive.org/web/20230801202928/https://www.science.org/content/blog-post/room-temperature-superconductor-new-developments>) from the original on 1 August 2023. Retrieved 1 August 2023 – via [Science.org](https://www.science.org).

27. Kim, Hyun-Tak (14 May 2021). "Room-temperature-superconducting Tc driven by electron correlation" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8121790>). *Scientific Reports*. **11** (1): 10329. Bibcode:2021NatSR..1110329K (<https://ui.adsabs.harvard.edu/abs/2021NatSR..1110329K>). doi:10.1038/s41598-021-88937-7 (<https://doi.org/10.1038/s41598-021-88937-7>). ISSN 2045-2322 (<https://www.worldcat.org/issn/2045-2322>). PMC 8121790 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8121790>). PMID 33990629 (<https://pubmed.ncbi.nlm.nih.gov/33990629>).
28. Brinkman, W. F.; Rice, T. M. (15 November 1970). "Application of Gutzwiller's Variational Method to the Metal-Insulator Transition" (<https://link.aps.org/doi/10.1103/PhysRevB.2.4302>). *Physical Review B*. **2** (10): 4302–4304. Bibcode:1970PhRvB...2.4302B (<https://ui.adsabs.harvard.edu/abs/1970PhRvB...2.4302B>). doi:10.1103/PhysRevB.2.4302 (<https://doi.org/10.1103/PhysRevB.2.4302>). ISSN 0556-2805 (<https://www.worldcat.org/issn/0556-2805>). Archived (<https://web.archive.org/web/20230802032758/https://journals.aps.org/prb/abstract/10.1103/PhysRevB.2.4302>) from the original on 2 August 2023. Retrieved 1 August 2023.
29. Hirsch, J. E. (23 January 1989). "Hole superconductivity" ([https://dx.doi.org/10.1016/0375-9601\(89\)90370-8](https://dx.doi.org/10.1016/0375-9601(89)90370-8)). *Physics Letters A*. **134** (7): 451–455. Bibcode:1989PhLA..134..451H (<https://ui.adsabs.harvard.edu/abs/1989PhLA..134..451H>). doi:10.1016/0375-9601(89)90370-8 ([https://doi.org/10.1016/0375-9601\(89\)90370-8](https://doi.org/10.1016/0375-9601(89)90370-8)). ISSN 0375-9601 (<https://www.worldcat.org/issn/0375-9601>). Archived (<https://web.archive.org/web/20140709222756/http://www.sciencedirect.com/science/article/pii/0375960189903708>) from the original on 9 July 2014. Retrieved 1 August 2023.
30. Griffin, Sinéad M. (31 July 2023). "Origin of correlated isolated flat bands in copper-substituted lead phosphate apatite". arXiv:2307.16892 (<https://arxiv.org/abs/2307.16892>) [cond-mat.supr-con ([https://arxiv.org/archive/cond-mat.supr-con](https://arxiv.org/archive/cond-mat/supr-con))].
31. Si, Liang; Held, Karsten (1 August 2023). "Electronic structure of the putative room-temperature superconductor **Pb₉Cu(PO₄)₆O**". *Physical Review B*. **108** (12). arXiv:2308.00676 (<https://arxiv.org/abs/2308.00676>). doi:10.1103/PhysRevB.108.L121110 (<https://doi.org/10.1103/PhysRevB.108.L121110>). S2CID 260351297 (<https://api.semanticscholar.org/CorpusID:260351297>).
32. Hao Wu; Li Yang; Bichen Xiao; Haixin Chang (3 August 2023). "Successful growth and room temperature ambient-pressure magnetic levitation of LK-99". arXiv:2308.01516 (<https://arxiv.org/abs/2308.01516>) [cond-mat.supr-con ([https://arxiv.org/archive/cond-mat.supr-con](https://arxiv.org/archive/cond-mat/supr-con))].
33. Padavic-Callaghan, Karmela (26 July 2023). "Room-temperature superconductor 'breakthrough' met with scepticism" (<https://archive.today/20230726181113/https://www.newscientist.com/article/2384782-room-temperature-superconductor-breakthrough-met-with-scepticism/#selection-689.37-689.118>). *New Scientist*. Archived from the original (<https://www.newscientist.com/article/2384782-room-temperature-superconductor-breakthrough-met-with-scepticism/>) on 26 July 2023. Retrieved 26 July 2023.
34. 关山口男子技师. 补充视频_哔哩哔哩_bilibili (<https://www.bilibili.com/video/BV13k4y1G7i1/>). *www.bilibili.com* (in Simplified Chinese). Archived (<https://web.archive.org/web/20230801154919/https://www.bilibili.com/video/BV13k4y1G7i1/>) from the original on 1 August 2023. Retrieved 1 August 2023.

35. 이병철; 최정석 (27 July 2023). '노벨상감' 상온 초전도체 세계 최초 개발했다는 한국 연구...과학계 '회의론' 넘을까 (<https://n.news.naver.com/article/366/0000920152>) [Korean study into world's first room-temperature superconductor ... can it overcome scientific 'skepticism' ... to win Nobel prize]. *Chosun Biz* (in Korean). Archived (<https://web.archive.org/web/20230727110149/https://n.news.naver.com/article/366/0000920152>) from the original on 27 July 2023. Retrieved 27 July 2023. 연구를 주도한 이석배 퀀텀에너지연구소 대표는 27일 오전 조선비즈와 만나 "2020년에 처음 연구 결과를 네이처에 제출했지만 다이어스 교수 사태 때문에 네이처가 논문 게재를 부담스러워했고, 다른 전문 학술지에 먼저 게재할 것을 요구했다"며 "국내 학술지에 먼저 올려서 국내 전문가의 검증을 받고 사전공개 사이트인 아카이브에 올린 것"이라고 말했다. 이 대표는 지난 23일 국제 학술지인 'ALP 머터리얼즈'에도 논문을 제출했다고 덧붙였다. 세계적인 물리학 저널에 인정을 받겠다는 설명이다. ... "지금은 작고한 최동식 고려대 화학과 교수와 함께 1990년대 중반부터 상온 초전도체 구현을 위해 20년에 걸쳐 연구와 실험을 진행했다"고 말했다. 이 대표는 상압 상온 초전도체에 대한 특허도 출원했다고 밝혔다.
36. Garisto, Dan (25 July 2023). "'A very disturbing picture': another retraction imminent for controversial physicist" (<https://www.nature.com/articles/d41586-023-02401-2>). *Nature*. **620** (7972): 14–16. Bibcode:2023Natur.620...14G (<https://ui.adsabs.harvard.edu/abs/2023Natur.620...14G>). doi:10.1038/d41586-023-02401-2 (<https://doi.org/10.1038%2Fd41586-023-02401-2>). PMID 37491414 (<https://pubmed.ncbi.nlm.nih.gov/37491414>). S2CID 260162594 (<https://api.semanticscholar.org/CorpusID:260162594>). Archived (<https://web.archive.org/web/20230727204107/https://www.nature.com/articles/d41586-023-02401-2>) from the original on 27 July 2023. Retrieved 28 July 2023.
37. KR 102404607 (<https://worldwide.espacenet.com/textdoc?DB=EPODOC&IDX=KR102404607>), Lee, Sukbae & Kim, Ji-Hoon, "초전도체를 포함하는 저저항 세라믹화합물의 제조방법 및 그 화합물 [Method of manufacturing ceramic composite with low resistance including superconductors and the composite thereof]", published 2021-05-31, assigned to Quantum Energy Research Institute Co. Ltd.
38. WO application 2023027536 (<https://worldwide.espacenet.com/textdoc?DB=EPODOC&IDX=WO2023027536>), Lee, Sukbae; Kim, Ji-Hoon & Kwon, Young-Wan, "상온, 상압 초전도 세라믹화합물 및 그 제조방법 [Room temperature and normal pressure superconducting ceramic compound, and method for manufacturing same]", published 2023-03-02, assigned to Quantum Energy Research Institute Co. Ltd.
39. "Room-temperature and atmospheric-pressure superconducting ceramic compound and preparation method therefor" (<https://patents.google.com/patent/WO2023027537A1/en>). 2 March 2023. Archived (<https://web.archive.org/web/20230727154613/https://patents.google.com/patent/WO2023027537A1/en>) from the original on 27 July 2023. Retrieved 5 August 2023.
40. LK-99 (http://engdtj.kipris.or.kr/engdtj/grrt1000a.do?method=biblioTMFrame&masterKey=4020230059989&index=0&kindOfReq=A&valid_fg=). *Korea Intellectual Property Rights Information Service* (Report). Korean Intellectual Property Office. 4 April 2023. Archived (http://web.archive.org/web/20230726200823/http://engdtj.kipris.or.kr/engdtj/grrt1000a.do?method=biblioTMFrame&masterKey=4020230059989&index=0&kindOfReq=A&valid_fg=) from the original on 26 July 2023. Retrieved 25 July 2023. "LK-99; ... Applicant: Quantum Energy Research Centre (Q-Centre); ... Status: Awaiting Examination"
41. Henshall, Will (3 August 2023). "Why Experts Are Skeptical About That Supposed Superconductor Breakthrough" (<https://time.com/6301391/experts-skeptical-about-ambient-superconductor/>). *Time*. Archived (<https://web.archive.org/web/20230803193547/https://time.com/6301391/experts-skeptical-about-ambient-superconductor/>) from the original on 3 August 2023. Retrieved 3 August 2023.
42. "Request for a verification sample from the Quantum Energy Research Institute... Receive a response 2 to 4 weeks after the thesis review" (<https://m.yna.co.kr/view/AKR20230803120200017>). *Yonhap News Agency*. 3 August 2023. Retrieved 3 August 2023.

43. Kwon, Young-Wan (28 July 2023). *The World First: Room-Temperature Ambient-Pressure Superconductor*. MML 2023: 11th International Symposium on Metallic Multilayers (<https://www.mml2023.org/>) (conference presentation). Korea University, Seoul, Korea: The Korean Magnetics Society.
44. Seifert, Tom S. [@TeraTom_S] (28 July 2023). "Just listening to an impressive talk of one of the coauthors of the room-temperature superconductor #LK99 at Korea university, Young-Wan Kwon" (https://x.com/TeraTom_S/status/1684808086547238912) (Tweet). Retrieved 28 July 2023 – via [Twitter](#).
45. Bodin, Kenneth [@KennethBodin] (28 July 2023). "They have now also presented at MML2023. They took questions. Answers not entirely satisfying. Rumour is that MIT, SC specialists are flying over to scrutinize experiments. (Photo @JohanaAkerman [Johaa Akerman])" (<https://x.com/KennethBodin/status/1684831376850157568>) (Tweet). Retrieved 28 July 2023 – via [Twitter](#).
46. "researcher from Q-Centre speaks with SBS News" (<https://mnews.sbs.co.kr/news/endPage.do?newsId=N1007295801&plink=SEARCH&cooper=SBSNEWSSEARCH>). 4 August 2023. Archived (<https://web.archive.org/web/20230804131746/https://news.sbs.co.kr/news/endPage.do?newsId=N1007295801&plink=SEARCH&cooper=SBSNEWSSEARCH>) from the original on 4 August 2023. Retrieved 4 August 2023.
47. Garisto, Dan (4 August 2023). "Claimed superconductor LK-99 is an online sensation — but replication efforts fall short" (<https://www.nature.com/articles/d41586-023-02481-0>). *Nature*. **620** (7973): 253. Bibcode:2023Natur.620..253G (<https://ui.adsabs.harvard.edu/abs/2023Natur.620..253G>). doi:10.1038/d41586-023-02481-0 (<https://doi.org/10.1038%2Fd41586-023-02481-0>). PMID 37542137 (<https://pubmed.ncbi.nlm.nih.gov/37542137>). S2CID 260544005 (<https://api.semanticscholar.org/CorpusID:260544005>). Archived (<https://web.archive.org/web/20230805012318/https://www.nature.com/articles/d41586-023-02481-0>) from the original on 5 August 2023. Retrieved 5 August 2023.
48. Kim, Jin-Won. Haeyoung Park (ed.). "S.Korean academics to verify truth of room-temperature superconductor" (<https://www.kedglobal.com/tech,-media-telecom/newsView/ked202308020020>). Tech, Media & Telecom. *The Korea Economic Daily Global Edition*. Archived (<https://web.archive.org/web/20230802153949/https://www.kedglobal.com/tech,-media-telecom/newsView/ked202308020020>) from the original on 2 August 2023. Retrieved 2 August 2023.
49. Ritchie, Stuart (26 July 2023). "The latest mega-breakthrough on room-temperature superconductors is probably nonsense" (<https://inews.co.uk/news/science/room-temperature-superconductors-latest-breakthrough-nonsense-2506774>). *i*. Archived (<https://web.archive.org/web/20230726220126/https://inews.co.uk/news/science/room-temperature-superconductors-latest-breakthrough-nonsense-2506774>) from the original on 26 July 2023. Retrieved 27 July 2023. "What about that levitation video? Dr Sven Friedemann, associate professor at the University of Bristol's School of Physics, told *i* that it, and other data in the paper, "could stem from other phenomena". Graphene, ... "is also diamagnetic [displaying repulsion like a superconductor] and can produce weak levitation". The video, in other words, could have a non-superconductor explanation."
50. Barber, Gregory (2 August 2023). "Inside the DIY Race to Replicate LK-99" (<https://www.wired.com/story/inside-the-diy-race-to-replicate-lk-99/>). *Wired*. Archived (<https://web.archive.org/web/20230802143413/https://www.wired.com/story/inside-the-diy-race-to-replicate-lk-99/>) from the original on 2 August 2023. Retrieved 2 August 2023.
51. Ferreira, Becky; Pearson, Jordan (1 August 2023). "DIY Scientists and Institutions Are Racing to Replicate the Room-Temperature Superconductor" (<https://www.vice.com/en/article/5d9yez/diy-scientists-and-institutions-are-racing-to-replicate-the-room-temperature-superconductor>). *Vice*. Archived (<https://web.archive.org/web/20230802153947/https://www.vice.com/en/article/5d9yez/diy-scientists-and-institutions-are-racing-to-replicate-the-room-temperature-superconductor>) from the original on 2 August 2023. Retrieved 2 August 2023.

52. Griffin, Andrew (27 July 2023). "Superconductor breakthrough could represent 'biggest physics discovery of a lifetime' – but scientists urge caution" (<https://www.independent.co.uk/tech/superconductor-breakthrough-electricity-power-paper-b2382711.html>). *www.independent.co.uk*. The Independent. Archived (<https://web.archive.org/web/20230728223136/https://www.independent.co.uk/tech/superconductor-breakthrough-electricity-power-paper-b2382711.html>) from the original on 28 July 2023. Retrieved 2 August 2023.
53. Pearson, Jordan (27 July 2023). "Viral Superconductor Study Claims to 'Open a New Era for Humankind.' Scientists Aren't So Sure" (<https://www.vice.com/en/article/bvj7pm/viral-room-temperature-superconductor-study-lk-99>). *www.vice.com*. Vice News. Archived (<https://web.archive.org/web/20230802154007/https://www.vice.com/en/article/bvj7pm/viral-room-temperature-superconductor-study-lk-99>) from the original on 2 August 2023. Retrieved 2 August 2023.
54. Written at Bloomberg News. "S Korea experts to test superconductor breakthrough claim" (<https://www.taipeitimes.com/News/biz/archives/2023/08/04/2003804195>). Taipei Times. 4 August 2023. Archived (<https://web.archive.org/web/20230804131756/https://www.taipeitimes.com/News/biz/archives/2023/08/04/2003804195>) from the original on 4 August 2023. Retrieved 4 August 2023.
55. Peng, Dannie (2 August 2023). "Superconductor breakthrough could represent 'biggest physics discovery of a lifetime' – but scientists urge caution" (<https://www.scmp.com/news/china/science/article/3229778/could-chinese-teams-viral-lk-99-video-offer-clue-superconductor-holy-grail-physicists>). *www.scmp.com*. South China Morning Post. Archived (<https://web.archive.org/web/20230802152233/https://www.scmp.com/news/china/science/article/3229778/could-chinese-teams-viral-lk-99-video-offer-clue-superconductor-holy-grail-physicists>) from the original on 2 August 2023. Retrieved 2 August 2023.
56. Griffin, Andrew (1 August 2023). "LK-99: Excitement rises over possibly revolutionary 'miracle material' – but there is still no good reason to believe it exists" (<https://www.independent.co.uk/tech/superconductor-lk-99-material-room-temperature-ambient-b2385781.html>). *www.independent.co.uk*. The Independent. Archived (<https://web.archive.org/web/20230802071250/https://www.independent.co.uk/tech/superconductor-lk-99-material-room-temperature-ambient-b2385781.html>) from the original on 2 August 2023. Retrieved 2 August 2023.
57. Ryan, Jackson (2 August 2023). "LK-99 Superconductor: Maybe a Breakthrough, Maybe Not So Much" (<https://web.archive.org/web/20230802181810/https://www.cnet.com/tech/computing/lk-99-superconductor-maybe-a-breakthrough-maybe-not-so-much/>). *www.cnet.com*. CNET. Archived from the original (<https://www.cnet.com/tech/computing/lk-99-superconductor-maybe-a-breakthrough-maybe-not-so-much/>) on 2 August 2023. Retrieved 2 August 2023.
58. "Superconductor LK-99 Breakthrough Buzz Spurs China, Korea Tech Rally" (<https://web.archive.org/web/20230802200014/https://www.bloomberg.com/news/articles/2023-08-02/superconductor-breakthrough-buzz-spurs-china-korea-tech-rally>). *www.bloomberg.com*. Bloomberg News. 1 August 2023. Archived from the original (<https://www.bloomberg.com/news/articles/2023-08-02/superconductor-breakthrough-buzz-spurs-china-korea-tech-rally>) on 2 August 2023. Retrieved 2 August 2023.
59. Culpan, Tim (2 August 2023). Written at Bloomberg News. "LK-99 and the Desperation for Scientific Discovery" (https://www.washingtonpost.com/business/energy/2023/08/02/lk-99-and-the-desperation-for-scientific-discovery/74c4f774-317a-11ee-85dd-5c3c97d6acda_story.html). *The Washington Post*. Washington Post. Archived (https://web.archive.org/web/20230803140836/https://www.washingtonpost.com/business/energy/2023/08/02/lk-99-and-the-desperation-for-scientific-discovery/74c4f774-317a-11ee-85dd-5c3c97d6acda_story.html) from the original on 3 August 2023. Retrieved 3 August 2023.

60. Nellis, Stephen (4 August 2023). "Superconductor claims spark investor frenzy, but scientists are skeptical" (<https://www.reuters.com/technology/superconductor-claims-spark-investor-frenzy-scientists-are-skeptical-2023-08-03/>). Reuters. Archived (<https://web.archive.org/web/20230804021045/https://www.reuters.com/technology/superconductor-claims-spark-investor-frenzy-scientists-are-skeptical-2023-08-03/>) from the original on 4 August 2023. Retrieved 4 August 2023.
61. Kang, Shinhye; Lee, Youkyung (9 August 2023). "Superconductor Stocks Drop in Korea Amid Doubts on Breakthrough" (<https://www.bloomberg.com/news/articles/2023-08-09/superconductor-stocks-drop-in-korea-amid-doubts-on-breakthrough>). Bloomberg News. Archived (<https://web.archive.org/web/20230810132925/https://www.bloomberg.com/news/articles/2023-08-09/superconductor-stocks-drop-in-korea-amid-doubts-on-breakthrough>) from the original on 10 August 2023. Retrieved 9 August 2023.
62. Kim, Jin-Won (17 August 2023). "Korea's superconductor stocks plunge on Nature report" (<https://www.kedglobal.com/tech,-media-telecom/newsView/ked202308170016>). The Korea Economic Daily. Archived (<https://web.archive.org/web/20230818001932/https://www.kedglobal.com/tech,-media-telecom/newsView/ked202308170016>) from the original on 18 August 2023. Retrieved 18 August 2023.
63. "Research on LK-99 Superconductor at Southeast University" (<https://targum.video/embed/388e770217a654ba9175b391eedd7802>). *targum.video*. Archived (<https://web.archive.org/web/20230802232401/https://targum.video/embed/388e770217a654ba9175b391eedd7802>) from the original on 2 August 2023. Retrieved 2 August 2023.
64. @condensed_the (3 August 2023). "Southeast may have drawn their figure misleadingly. On a linear scale, there seems to be no transition, very disappointing and not a good sign since the artifact also looms large" (https://x.com/condensed_the/status/1686895266329174016) (Tweet). Retrieved 3 August 2023 – via Twitter.
65. P. Puphal; M. Y. P. Akbar; M Hepting; E. Goering; M. Isobe; A. A. Nugroho; B. Keimer (11 August 2023). "Single crystal synthesis, structure, and magnetism of $\text{Pb}_{10-x}\text{Cu}_x(\text{PO}_4)_6\text{O}$ ". *APL Materials*. **11** (10). arXiv:2308.06256 (<https://arxiv.org/abs/2308.06256>). Bibcode:2023APLM...11j1128P (<https://ui.adsabs.harvard.edu/abs/2023APLM...11j1128P>). doi:10.1063/5.0172755 (<https://doi.org/10.1063%2F5.0172755>). S2CID 260866146 (<https://api.semanticscholar.org/CorpusID:260866146>).
66. Knapp, Alex (18 August 2023). "The Rise And Fall Of The Would-Be Superconductor That Transfixed The Internet" (<https://www.forbes.com/sites/alexknapp/2023/08/18/the-rise-and-fall-of-the-would-be-superconductor-that-transfixed-the-internet>). *Forbes*. Archived (<https://web.archive.org/web/20230818123335/https://www.forbes.com/sites/alexknapp/2023/08/18/the-rise-and-fall-of-the-would-be-superconductor-that-transfixed-the-internet>) from the original on 18 August 2023. Retrieved 18 August 2023.
67. 关山口男子技师. "LK-99验证_哔哩哔哩_bilibili" (<https://www.bilibili.com/video/BV14p4y1V7kS/>). *www.bilibili.com* (in Simplified Chinese). Archived (<https://web.archive.org/web/20230801090615/https://www.bilibili.com/video/BV14p4y1V7kS/>) from the original on 1 August 2023. Retrieved 1 August 2023.
68. Li Liu; Ziang Meng; Xiaoning Wang; Hongyu Chen; Zhiyuan Duan; Xiaorong Zhou; Han Yan; Peixin Qin; Zhiqi Liu (31 July 2023). "Semiconducting Transport in $\text{Pb}_{10-x}\text{Cu}_x(\text{PO}_4)_6\text{O}$ Sintered from Pb_2SO_5 and Cu_3P ". *Advanced Functional Materials*. arXiv:2307.16802 (<https://arxiv.org/abs/2307.16802>). doi:10.1002/adfm.202308938 (<https://doi.org/10.1002%2Fadfm.202308938>). S2CID 260334279 (<https://api.semanticscholar.org/CorpusID:260334279>).

69. Tran, Tony Ho (1 August 2023). "Sorry, But the New LK-99 Superconductor Breakthrough Might Be Total BS" (<https://www.thedailybeast.com/why-the-room-temperature-lk-99-superconductor-might-be-total-bs>). *www.thedailybeast.com*. The Daily Beast. Archived (<https://web.archive.org/web/20230801224658/https://www.thedailybeast.com/why-the-room-temperature-lk-99-superconductor-might-be-total-bs>) from the original on 1 August 2023. Retrieved 2 August 2023.
70. Hou, Qiang; Wei, Wei; Zhou, Xin; Sun, Yue; Shi, Zhixiang (2 August 2023). "Synthesis, transport and magnetic properties of Cu-doped apatite $\text{Pb}_{10}\text{-Cu}(\text{PO}_4)_6\text{O}$ ". *Matter*. **6** (12): 4408–4418. [arXiv:2308.01192](https://arxiv.org/abs/2308.01192) (<https://arxiv.org/abs/2308.01192>). doi:10.1016/j.matt.2023.11.014 (<https://doi.org/10.1016%2Fj.matt.2023.11.014>).
71. Hou, Qiang; Wei, Wei; Zhou, Xin; Wang, Xinyue; Sun, Yue; Shi, Zhixiang (2 August 2023). "Synthesis, transport and magnetic properties of Cu-doped apatite $\text{Pb}_{10}\text{-Cu}(\text{PO}_4)_6\text{O}$ ". *Matter*. **6** (12): 4408–4418. [arXiv:2308.05778](https://arxiv.org/abs/2308.05778) (<https://arxiv.org/abs/2308.05778>). doi:10.1016/j.matt.2023.11.014 (<https://doi.org/10.1016%2Fj.matt.2023.11.014>).
72. Kaizhen Guo; Yuan Li; Shuang Kia (6 August 2023). "Ferromagnetic half levitation of LK-99-like synthetic samples". *Science China Physics, Mechanics & Astronomy*. **66** (10). [arXiv:2308.03110](https://arxiv.org/abs/2308.03110) (<https://arxiv.org/abs/2308.03110>). Bibcode:2023SCPMA..6607411G (<https://ui.adsabs.harvard.edu/abs/2023SCPMA..6607411G>). doi:10.1007/s11433-023-2201-9 (<https://doi.org/10.1007%2Fs11433-023-2201-9>). S2CID 260680385 (<https://api.semanticscholar.org/CorpusID:260680385>).
73. Jicheng Liu; Chenao He; et al. (Weijie Huang, Zhihao Zhen, Guanhua Chen, Tianyong Luo, Xianfeng Qiao, Yao Yao, Dongge Ma) (16 December 2023). "Strange memory effect of low-field microwave absorption in copper-substituted lead apatite". [arXiv:2312.10391](https://arxiv.org/abs/2312.10391) (<https://arxiv.org/abs/2312.10391>) [cond-mat (<https://arxiv.org/archive/cond-mat>)].
74. Jiang, Yi; Lee, Scott B.; Herzog-Arbeitman, Jonah; Yu, Jiabin; Feng, Xiaolong; Hu, Haoyu; Călugăru, Dumitru; Brodale, Parker S.; Gormley, Eoghan L.; Maia Garcia Vergniory; Felser, Claudia; Blanco-Canosa, S.; Hendon, Christopher H.; Schoop, Leslie M.; Andrei Bernevig, B. (8 August 2023). "Pb₉Cu(PO₄)₆(OH)₂: Phonon bands, Localized Flat Band Magnetism, Models, and Chemical Analysis". [arXiv:2308.05143](https://arxiv.org/abs/2308.05143) (<https://arxiv.org/abs/2308.05143>) [cond-mat.supr-con (<https://arxiv.org/archive/cond-mat.supr-con>)].
75. Timokhin, Ivan; Chen, Chuhongxu; Yang, Qian; Mishchenko, Artem (7 August 2023). "Synthesis and characterisation of LK-99". [arXiv:2308.03823](https://arxiv.org/abs/2308.03823) (<https://arxiv.org/abs/2308.03823>) [cond-mat.supr-con (<https://arxiv.org/archive/cond-mat.supr-con>)].
76. V.P.S Awana (10 August 2023). "LK-99 VIDEO" (<https://www.facebook.com/100008122782686/videos/661722942350287>). Facebook. Archived (<https://web.archive.org/web/2023081012413/https://www.facebook.com/100008122782686/videos/661722942350287/>) from the original on 10 August 2023. Retrieved 10 August 2023.
77. "People@CSIR-NPL – NPL" (<https://www.nplindia.org/index.php/peoplecsir-npl/>). Archived (<https://web.archive.org/web/20230608193251/https://www.nplindia.org/index.php/peoplecsir-npl/>) from the original on 8 June 2023. Retrieved 31 July 2023.
78. "Dr. V.P.S. Awana, PhD - Editorial Board - Superconductivity - Journal - Elsevier" (<https://www.journals.elsevier.com/superconductivity/editorial-board/dr-vps-awana-phd>). *www.journals.elsevier.com*. Archived (<https://web.archive.org/web/20230803191711/https://www.journals.elsevier.com/superconductivity/editorial-board/dr-vps-awana-phd>) from the original on 3 August 2023. Retrieved 31 July 2023.
79. @andrewmccalip (4 August 2023). "Meissner effect or bust: Day 8.5" (<https://x.com/andrewmccalip/status/1687405505604734978>) (Tweet). Retrieved 4 August 2023 – via Twitter.
80. @andrewmccalip (6 August 2023). "[untitled]" (<https://x.com/andrewmccalip/status/1688082372250755072>) (Tweet). Retrieved 6 August 2023 – via Twitter.
81. @andrewmccalip (31 July 2023). "Meissner effect or bust: Day 4" (<https://x.com/andrewmccalip/status/1685871360948748288>) (Tweet). Retrieved 1 August 2023 – via Twitter.

82. Cho, Adrian (27 July 2023). "A spectacular superconductor claim is making news. Here's why experts are doubtful" (<https://www.science.org/content/article/spectacular-superconduct-or-claim-making-news-here-s-why-experts-are-doubtful>). *Science*. American Association for the Advancement of Science. doi:10.1126/science.adk0021 (<https://doi.org/10.1126%2Fscience.adk0021>). Archived (<https://web.archive.org/web/20230729055410/https://www.science.org/content/article/spectacular-superconductor-claim-making-news-here-s-why-experts-are-doubtful>) from the original on 29 July 2023. Retrieved 29 July 2023. "Michael Norman, a theorist at Argonne National Laboratory ... says, researchers at Argonne and elsewhere are already trying to replicate the experiment."
83. "科技周报|国内发现疑似室温超导新材料、比亚迪销量超特斯拉_腾讯新闻" (<https://new.qq.com/rain/a/20240105A088TQ00>). *new.qq.com*. Retrieved 8 January 2024.
84. "又导了？中科院等发现新疑似室温超导材料，作者：对结果很有信心_澎湃号·湃客_澎湃新闻-The Paper" (https://www.thepaper.cn/newsDetail_forward_25898448). *www.thepaper.cn*. Retrieved 8 January 2024.
85. Hongyang Wang; Yao Yao; et al. (Ke Shi, Yijing Zhao, Hao Wu, Zhixing Wu, Zhihui Geng, Shufeng Ye, Ning Chen) (2 January 2024). "Possible Meissner effect near room temperature in copper-substituted lead apatite". *arXiv:2401.00999* (<https://arxiv.org/abs/2401.00999>).
86. Hongyang Wang (3 January 2024). "如何看待「真可爱呆」等人论文(2401.00999)发现可能的近室温迈斯纳效应（疑似室温超导）？” (<https://www.zhihu.com/question/637763289/answer/3347898701>) [How to view the possible near room temperature Meisner effect (suspected of room temperature superconductivity) discovered in the paper by "Truly Cute and Stupid" and others (2401.00999)?]. *Zhihu* (in Chinese (China)). Archived (<https://archive.today/20240103082305/https://www.zhihu.com/question/637763289/answer/3347898701>) from the original on 3 January 2024. Retrieved 16 January 2024.
87. Yao Yao (3 January 2024). "如何看待「真可爱呆」等人论文(2401.00999)发现可能的近室温迈斯纳效应（疑似室温超导）？” (<https://www.zhihu.com/question/637763289/answer/3347895981>) [How to view the possible near room temperature Meisner effect (suspected of room temperature superconductivity) discovered in the paper by "Truly Cute and Stupid" and others (2401.00999)?]. *Zhihu* (in Chinese (China)). Archived (<https://archive.today/20240103082258/https://www.zhihu.com/question/637763289/answer/3347895981>) from the original on 3 January 2024. Retrieved 16 January 2024.
88. Zhihui Geng (3 January 2024). "如何看待「真可爱呆」等人论文(2401.00999)发现可能的近室温迈斯纳效应（疑似室温超导）？” (<https://www.zhihu.com/question/637763289/answer/3347887710>) [How to view the possible near room temperature Meisner effect (suspected of room temperature superconductivity) discovered in the paper by "Truly Cute and Stupid" and others (2401.00999)?]. *Zhihu* (in Chinese (China)). Archived (<https://archive.today/20240103082249/https://www.zhihu.com/question/637763289/answer/3347887710>) from the original on 3 January 2024. Retrieved 16 January 2024.
89. Lai, Junwen; Li, Jiangxu; Liu, Peitao; Sun, Yan; Chen, Xing-Qiu (29 July 2023). "First-principles study on the electronic structure of $\text{Pb}_{10-x}\text{Cu}_x(\text{PO}_4)_6\text{O}$ ($x = 0, 1$)". *Journal of Materials Science & Technology*. **171**: 66–70. arXiv:2307.16040 (<https://arxiv.org/abs/2307.16040>). doi:10.1016/j.jmst.2023.08.001 (<https://doi.org/10.1016%2Fj.jmst.2023.08.001>).
90. "Breakthrough in Superconductivity: Huazhong University Scientists Report First Successful Replication of LK-99" (<https://beijingtimes.com/china/2023/08/01/breakthrough-in-superconductivity-huazhong-university-scientists-report-first-successful-replication-of-lk-99/>). *Beijing Times*. 1 August 2023. Archived (<https://web.archive.org/web/20230802153951/https://beijingtimes.com/china/2023/08/01/breakthrough-in-superconductivity-huazhong-university-scientists-report-first-successful-replication-of-lk-99/>) from the original on 2 August 2023. Retrieved 2 August 2023.

91. Kurlito, Rafal; Lany, Stephan; Pashov, Dimitar; Acharya, Swagata; van Schilfgaarde, Mark; Dessau, Daniel S. (1 August 2023). "Pb-apatite framework as a generator of novel flat-band CuO based physics, including possible room temperature superconductivity". [arXiv:2308.00698](https://arxiv.org/abs/2308.00698) (<https://arxiv.org/abs/2308.00698>) [cond-mat.supr-con (<https://arxiv.org/archive/cond-mat/supr-con>)].
92. Sinéad Griffin [@sineatrix] (2 August 2023). "a monster thread on what my paper says, the approximations and the caveats" (<https://x.com/sineatrix/status/1686659102674751488>) (Tweet). Retrieved 3 August 2023 – via [Twitter](#).
93. Condensed Matter Theory Center, UMD [@condensed_the] (1 August 2023). "For such flat band systems, packaged LDA type calculations are of limited utility, but knowing the LDA band structure is again a small, but necessary, step in understanding the physics. Flat bands DO NOT imply SC, flat bands often lead to magnetic instabilities" (https://x.com/condensed_the/status/1686373899662213123) (Tweet). Retrieved 2 August 2023 – via [Twitter](#).
94. Cabezas-Escases, J (2023). "Theoretical insight on the LK-99 material". [arXiv:2308.01135](https://arxiv.org/abs/2308.01135) (<https://arxiv.org/abs/2308.01135>) [cond-mat.supr-con (<https://arxiv.org/archive/cond-mat/supr-con>)].
95. Georgescu, Alexandru B. (2023). "Cu-doped $\text{Pb}_{10}(\text{PO}_4)_6\text{O}$, and V doped SrTiO_3 -- a tutorial on electron-crystal lattice coupling in insulating materials with transition metal dopants". [arXiv:2308.07295](https://arxiv.org/abs/2308.07295) (<https://arxiv.org/abs/2308.07295>) [cond-mat.str-el (<https://arxiv.org/archive/cond-mat/str-el>)].
96. @AlexandruBG (10 August 2023). "So here's some results on LK-99 as a tutorial case example in electron-lattice interactions. So here's my two main results on it. Just very isolated, $S=1/2$, Cu bands in an insulator. Nothing at the Fermi level in DFT+U. One can also think of these as color centers" (<https://x.com/AlexandruBG/status/1689735667041861633>) (Tweet). Retrieved 28 August 2023 – via [Twitter](#).
97. @AlexandruBG (16 August 2023). "Nice. Theory can be pretty predictive at times" (<https://x.com/AlexandruBG/status/1691876959737762172>) (Tweet). Retrieved 28 August 2023 – via [Twitter](#).

Further reading

- 최동식 (17 May 1994). [초전도 혁명의 이론적 체계](https://kupress.com/books/3248/) (<https://kupress.com/books/3248/>) [*Theoretical Framework of the Superconducting Revolution*] (in Korean). 고려대학교출판부. ISBN 89-7641-276-1. Archived (<https://web.archive.org/web/20230727055743/https://kupress.com/books/3248/>) from the original on 27 July 2023. Retrieved 27 July 2023.
- "최동식" (<https://dl.dongascience.com/magazine/view/S199309N024>). *Donga Science* (Interview with T.S. Chair) (in Korean): 106–107. September 1993. Archived (<https://web.archive.org/web/20230320130141/https://dl.dongascience.com/magazine/view/S199309N024>) from the original on 20 March 2023. Retrieved 3 August 2023.
- Krivovichev, Sergey V.; Burns, Peter C. (17 October 2002). "Crystal chemistry of lead oxide phosphates: crystal structures of $\text{Pb}_4\text{O}(\text{PO}_4)_2$, $\text{Pb}_8\text{O}_5(\text{PO}_4)_2$ and $\text{Pb}_{10}(\text{PO}_4)_6\text{O}$ ". *Zeitschrift für Kristallographie – Crystalline Materials*. **218** (5). Munich: Oldenbourg Wissenschaftsverlag (published 1 May 2003): 357–365. doi:10.1524/zkri.218.5.357.20732 (<https://doi.org/10.1524%2Fzkri.218.5.357.20732>). S2CID 101229927 (<https://api.semanticscholar.org/CorpusID:101229927>). "The compound $\text{Pb}_{10}(\text{PO}_4)_6\text{O}$ has been designated 'oxyppyromorphite' ... $\text{Pb}_{10}(\text{PO}_4)_6\text{O}$ crystallizes with an apatite-type structure. The structure contains a single O atom that is not part of a PO_4 tetrahedron; it has a site occupancy factor of 0.25 and is located on the 6_3 axis."
- Lowe, Derek (26 July 2023). "Breaking Superconductor News" (<https://www.science.org/content/blog-post/breaking-superconductor-news>). Chemical News. *In the Pipeline* (blog).

American Association for the Advancement of Science. Archived (<https://web.archive.org/web/20230726174644/https://www.science.org/content/blog-post/breaking-superconductor-new-s>) from the original on 26 July 2023. Retrieved 26 July 2023 – via Science.org.

- Orf, Darren (27 July 2023). "Scientists Claim They Found the Holy Grail of Superconductors" (<https://www.popularmechanics.com/science/energy/a44657321/room-temperature-superconductor/>). *Energy. Popular Mechanics*. Retrieved 28 July 2023.
- Harris, Margaret (27 July 2023). "Have scientists in Korea discovered the first room-temperature, ambient-pressure superconductor?" (<https://physicsworld.com/a/have-scientists-in-korea-discovered-the-first-room-temperature-ambient-pressure-superconductor/>). *Superconductivity blog*. Physics World. Institute of Physics. Archived (<https://web.archive.org/web/20230728233445/https://physicsworld.com/a/have-scientists-in-korea-discovered-the-first-room-temperature-ambient-pressure-superconductor/>) from the original on 28 July 2023. Retrieved 29 July 2023.
- Wu, Yuewei; Sun, Wenya; Li, Ruiyang (28 July 2023). 南大教授谈韩国室温超导：不像超导，正重复实验—新闻 (<https://news.sciencenet.cn/htmlnews/2023/7/505675.shtm>) [NU professor talks about room temperature superconductivity found in South Korea: Unlike superconductivity, experiments are being repeated]. *ScienceNet.cn* (broad overview) (in Chinese). Archived (<https://web.archive.org/web/20230731054842/https://news.sciencenet.cn/htmlnews/2023/7/505675.shtm>) from the original on 31 July 2023. Retrieved 31 July 2023.
- 안경애 (8 August 2023). "'LK-99 샘플' 미세 결정구조 논문과 같다"...에너지공대 확인 (https://m.dt.co.kr/contents.html?article_no=2023080802109931650002) [The 'LK-99' sample crystal structure matches published paper — [Korea University] College of Energy Engineering]. *Digital Times* (in Korean). Archived (https://web.archive.org/web/20230810132840/https://m.dt.co.kr/contents.html?article_no=2023080802109931650002) from the original on 10 August 2023. Retrieved 9 August 2023. "LK-99 연구에 참여하고 있는 한국에너지공대의 한 연구자는 8일 디지털타임스와의 통화에서 "샘플에 대한 X선 회절구조 분석 결과, 논문에 제시된 것과 샘플의 미세 결정구조가 같다는 사실을 확인했다"고 밝혔다. 이 연구자는 "미세 결정구조를 보면 물질의 물성을 확인하지 못해도 가능성을 추정할 수 있다. LK-99 분석을 위해 X레이로 미세 결정구조를 확인한 결과, 쿼텀에너지연구소 측이 논문에서 밝힌 결정구조와 샘플에서 확인되는 결정구조가 같았다"고 말했다."
- Shilin Zhu; Wei Wu; Zheng Li; Jianlin Luo (8 August 2023). "First-order transition in LK-99 containing Cu₂S". *Matter*. **6** (12): 4401–4407. arXiv:2308.04353 (<https://arxiv.org/abs/2308.04353>). doi:10.1016/j.matt.2023.11.001 (<https://doi.org/10.1016%2Fj.matt.2023.11.001>).
- Puphal, P.; Akbar, M. Y. P.; Hepting, M.; Goering, E.; Isobe, M.; Nugroho, A. A.; Keimer, B. (11 August 2023). "Single crystal synthesis, structure, and magnetism of Pb_{10-x}Cu_x(PO₄)₆O". *APL Materials*. **11** (10). arXiv:2308.06256 (<https://arxiv.org/abs/2308.06256>). Bibcode:2023APLM...11j1128P (<https://ui.adsabs.harvard.edu/abs/2023APLM...11j1128P>). doi:10.1063/5.0172755 (<https://doi.org/10.1063%2F5.0172755>). S2CID 260866146 (<https://api.semanticscholar.org/CorpusID:260866146>).

External links

- *Magnetic Property Test of LK-99 Film* (<https://www.youtube.com/watch?v=EtVjGWpbE7k>) (video). Quantum Energy Research Centre. 26 January 2023. Retrieved 25 July 2023 – via Youtube. This effect is a conductive copper plate induced by a magnetic.
- Kim, Hyun-Tak (25 July 2023). *Superconductor Pb_{10-x}Cu_x(PO₄)₆O showing levitation at room temperature and atmospheric pressure and mechanism* (<https://sciencecast.org/casts/suc384jly50n>) (video). Retrieved 25 July 2023 – via ScienceCast.
- List of replication claims, regularly updated:

- [Compilation of Known Replication Attempt Claims \(https://forums.spacebattles.com/threads/claims-of-room-temperature-and-ambient-pressure-superconductor.1106083/\)](https://forums.spacebattles.com/threads/claims-of-room-temperature-and-ambient-pressure-superconductor.1106083/). Guderian2nd, Spacebattles Forums. Retrieved 2 August 2023.
- [LK-99#Online Claims \(https://eirifu.wordpress.com/2023/07/30/lk-99-superconductor-summary/\)](https://eirifu.wordpress.com/2023/07/30/lk-99-superconductor-summary/). Eiri Sanada. Retrieved 2 August 2023.

Retrieved from "<https://en.wikipedia.org/w/index.php?title=LK-99&oldid=1223576476>"

■